LabSkin
SKIN SCIENCE

LabSkin full thickness

WITH A375 MELANOMA CELLS AS A MEDIUM TO STUDY MELANOMA

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LabSkin is a 3D full-thickness skin equivalent built on a fibrin gel. The human dermal adult fibroblasts (HDFa) are cast into the fibrin gel and cultured for six days. During this time the fibroblasts remodel the gel with ECM proteins. After the dermal culture period, the models are seeded with human epidermal neonatal keratinocytes (HEKn), cultured for two days submerged, raised to the ALI and cultured for a further 12 days. To introduce melanoma into the model, A375 were cultured in 96-well Haem- coated plates to form spheroid clusters of up to 1 x 10^5 cells. These cluster were then transferred to LabSkin-S prior to HEKn seeding, allowed to attach and then continued with the rest of LabSkin-S. protocol.

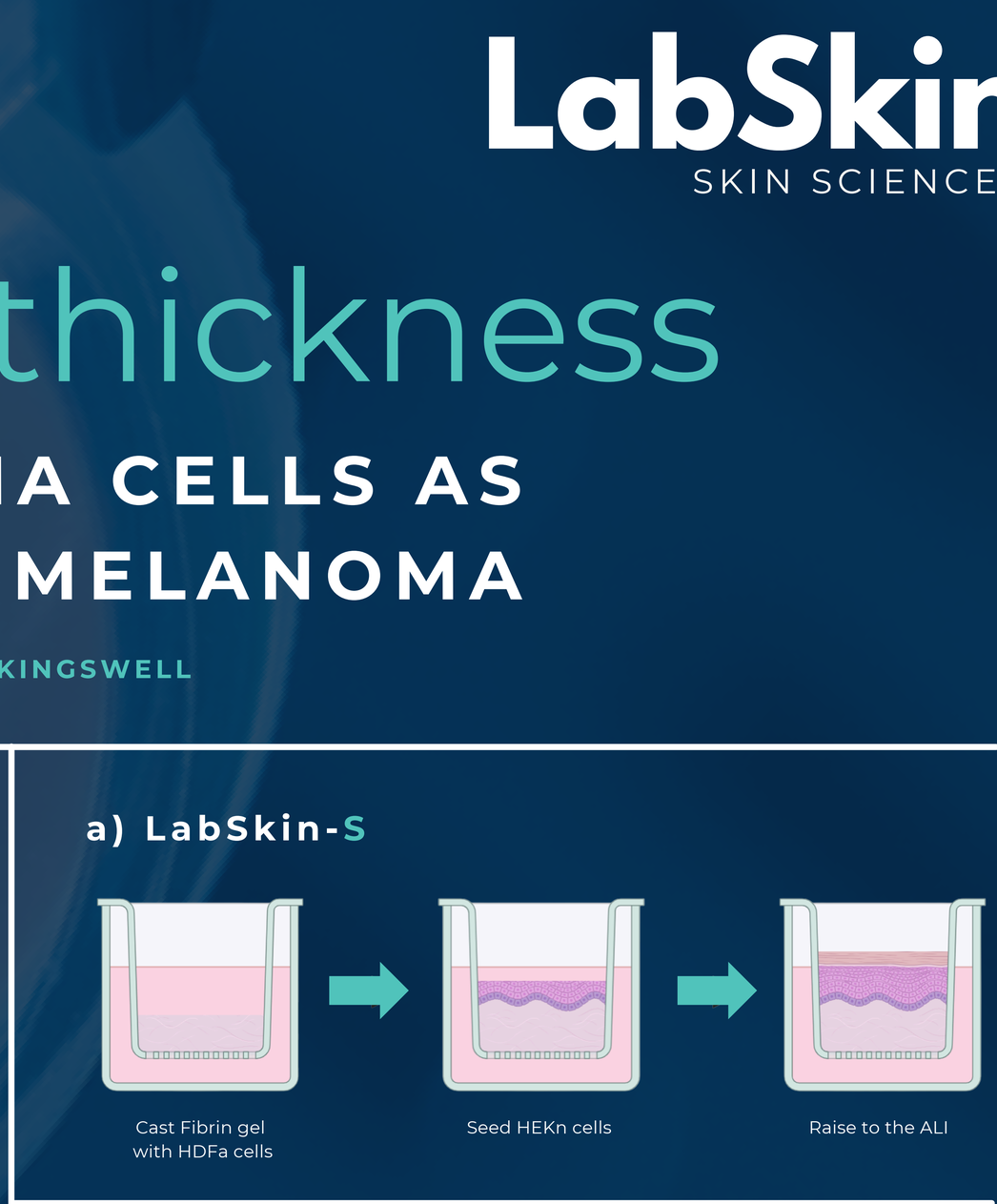


Figure 1. (a) A visual representation of the stages of LabSkin-S production. (b) The culture of melanoma cells in 96-well plates before the addition of melanoma cells prior to epidermal seeding.

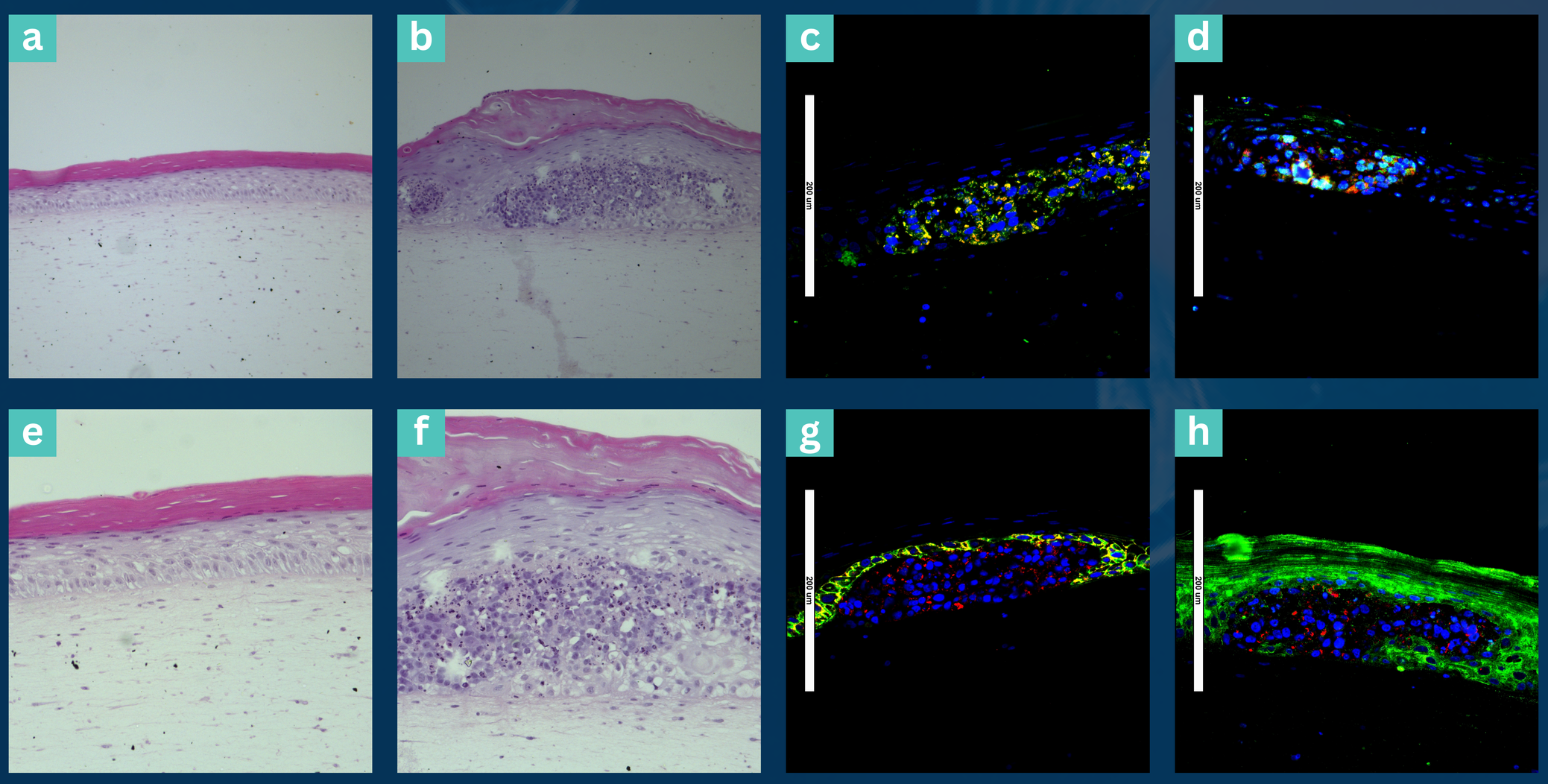


Figure 2. A side-by-side comparison of H&E staining of LabSkin-S 10x (a) and 20x (c) compared to LabSkin with A375 melanoma cells 10x (b) and 20x (d). Immunofluorescent staining of LabSkin with A375 melanoma cells stained against (e) ATPase (green) and Melanoma associated antigen (MAA) (red), (f) GP100 (green) and MAA (red), (g) Integrin (green) and CD4 (red), (h) Pan-Cytokeratin (green) and MAA (red).

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